

Privacy-Preserving Natural Language Processing

RUHR
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Lecture 10 : Real-World DP Applications and the Epsilon Registry

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CENTER FOR TRUSTWORTHY
DATA SCIENCE AND SECURITY

Tech Giants Applying DP

- 1 Tech Giants Applying DP
- 2 Other Organizations Applying DP
- 3 Standardizations and the Epsilon Registry

Apple User Data from macOS and iOS

Tons of user data being collected by Apple:

- Typing information for QuickType suggestions
- Emojis for emoji suggestions
- Web navigation data from Safari

Apple uses local DP with different ϵ values to protect user privacy in those data.

Full details are reported *here*.

Meta and Facebook User Interactions

DVN/TDOAPG_2020

Dataset of user interactions with web pages shared on FB is publicly available.

Examples:

- John shared URL foo
- Mary views post with URL bar

Meta uses (ϵ, δ) DP to protect each interaction.

Microsoft and Windows Telemetry

ding2017collecting

User data collections starting (massively) with Windows 10.

Examples:

- Application crash reports.
- User time spent on certain apps.
- Typing, location, connected devices, etc.

Microsoft uses local DP with $\epsilon = 1.67$ to protect it before using / sharing / selling.

Google Shopping and Trends

Product page view counts for prioritizing page crawling.

- Collected from users on daily basis.
- Streammed using DP-SQLP
- Protected using (ϵ, δ) DP with $\epsilon = 1$ and $\delta = 10^{-9}$

Search trends used to show related queries.

- Streammed using DP-SQLP
- Protected using (ϵ, δ) DP with $\epsilon = 2$ and $\delta = 10^{-10}$

Google Gboard next-word prediction

xu2023federatedlearninggboardlanguage

Federated learning and DP to build next-word prediction models for Gboard, a virtual keyboard for android.

- Each model uses (ϵ, δ) DP with:
 - ϵ from 0.69 to 10.61
 - $\delta = 10^{-5}$
- Models trained with TensorFlow Federated¹ and TensorFlow Privacy²

¹<https://www.tensorflow.org/federated>

²https://www.tensorflow.org/responsible_ai/privacy

Gboard out-of-vocabulary word discovery

sun2024privatefederateddiscoveryoutofvo

Distributed DP for discovering new words (out-of-vocabulary) typed by users in Gboard, and enhance the performance of the keyboard.

- Local models using ϵ DP with $\epsilon = 10$
- Central guarantee (ϵ, δ) DP with $\epsilon = 0.32$ and $\delta = 10^{-10}$

Full details are reported *here*.

Linkedin Audience Engagement

DBLP:journals/corr/abs-2002-05839

Interactive query system that allows marketers to get information about LinkedIn users engaging with their content.

- Each analyst sends queries to the system
- Each query returns (ϵ, δ) DP with $\epsilon = 0.15$ and $\delta = 10^{-10}$
- Monthly limit (budget) for queries is $\epsilon = 34.9$ and $\delta = 7 \cdot 10^{-9}$
- Extra measures to prevent averaging attacks

Labor Market Insights

rogers2020membersapproachenablinglink

Data from LinkedIn to measure trends in people changing their occupation.

- Companies who are hiring, using with $\varepsilon = 14.4$ and $\delta = 1.2 \cdot 10^{-9}$
- Available jobs, protecting hiring event with $\varepsilon = 14.4$ and $\delta = 1.2 \cdot 10^{-9}$
- Required skills, protecting each user's skills information with $\varepsilon = 0.3$ and $\delta = 3 \cdot 10^{-10}$
- Extra measures to prevent averaging attacks

Other Organizations Applying DP

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USA Census Bureau

County Business Patterns

- Business establishments in the USA
- Different types of data (including finances)
- Protected with various (ϵ, δ) parameters based on:
 - Financial assets
 - Annual payroll
 - Number of employs

Full details are reported *here*.

The 2020 USA Census

- Collecting demographic information about USA population
- Datasets protected with various (ϵ, δ) parameters
 - *Redistricting* data protected with $\epsilon = 13.64$
 - *Demographic Housing* protected with $\epsilon = 19.46$
 - Racial and ethnical categories protected with $\epsilon = 45.68$
 - Supplemental household statistics protected with $\epsilon = 12.74$

Post-Secondary Employment Outcomes

- Earnings and employment of college graduates
- Categorized by degree level, degree major, and post-secondary institution
- Matching university transcript data with national database of jobs
- Each person protected with $\epsilon = 1.5$

Full details are reported *here*.

Page view statistics

- Distinct users visiting Wiki pages each day from each country
- July 1, 2015 to Feb. 8, 2017 protected with $\varepsilon = 1$ on 300 pages view per day
- Feb. 9, 2017 to Feb. 5, 2023 protected with $\varepsilon = 1$ on 30 pages view per day
- Feb. 6, 2023 onwards protected with $\varepsilon = 0.72$ and $\delta = 10^{-5}$

Editor statistics

- Statistics about editor activity by project and country
- Monthly published data protected with $\varepsilon = 2$ on editor-project-country-month
- Weekly published data protected with $\varepsilon = 2$ on editor-project-country-week
- One-off release for Russian editors, protected with $\varepsilon = 0.1$

Other Applications

- Brave uses distributed DP for collecting user analytics
- Spectus published a dashboard with mobility trends during Hurricane Irma
- Microsoft Assistive AI collects user data about Office Tools and suggests automatic replies
- Google uses DP and federated learning for collecting data and training models for improved text selection and copying on Android

Standardizations and the Epsilon Registry

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Background: Evolution of Information Security

Early days, until the '60s

- Mostly physical security: focus on access to mainframes
- Limited threats: computers were rare and isolated
- Computers were mostly used for scientific computations; little or no personal data involved
- Basic encryption: simple ciphers like Caesar cipher were used to ensure message confidentiality

Background: Evolution of Information Security

Rise of the networks, '60s - '90s

- Computer networks and remote access become common
- Unauthorized network / system access and data breaches become popular concerns
- Cybersecurity emerges as an important discipline
- Many remote access protocols with encryption introduced
- Popular symmetric cyphers of 64 bits like DES
- Public key cryptography comes out

Background: Evolution of Information Security

The internert era, '00s - present

- Widespread internet usage, online shopping, personal data sharing
- Surge in cybercrime, viruses, worms, ransomware, phishing, DDOS attacks
- Evolution of security measures: VPNs, firewalls, antiviruses, intrusion detection systems
- Evolution of cryptography: 128 bit cyphers like AES become mandatory

Information Security Standards

- **ISO/IEC 27001:** Standards for establishing, implementing, maintaining and improving an ISMS
- **ISO/IEC 27002:** provides a code of practice for information security controls, offering guidance for implementing security policies
- **ISO/IEC 27005:** focuses on information security risk management, helping organizations in assessing and managing risks

Information Privacy vs. Security

- Privacy is inherently a more complex concept: who, what, when, where, why
- Private information is highly context dependent which makes it difficult to protect
- Lack of motivation (unless enforced by law) to protect privacy because of utility (profitability) loss
- Can privacy evolution and standardization follow the same path as security?

The Epsilon Registry

dwork_differential_2019

- DP is being successfully implemented in industry, public sector, academia, etc.
- Still, little understanding of the optimal ε values for certain systems, purposes, data types, etc.
- No clear consensus among practitioners about:
 - How to approach the key implementation decisions
 - How to decide about the right privacy guarantees
 - How to choose ε levels

Could a public Epsilon Registry help?

- Communal body of knowledge about DP implementations
- Used by stakeholders, public servants, academics, etc.
- Serve as guideline for identifying and adopting judicious DP implementations

Questions...?

THANK YOU...!